

CoRoT-1b

R-2491

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_210202 · created 2026-08-24T01:17:43+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459247.6915 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.043
Transit depth (Rp/Rs)^2	2.03 +/- 1.22 [%]
Orbital Inclination (inc)	89.61 +/- 2.81
Airmass coefficient 1 (a1)	0.273 +/- 0.095
Airmass coefficient 2 (a2)	0.69 +/- 0.48
Scatter in the residuals of the lightcurve fit is	39.43 %
Best Comparison Star	#3 - [197, 100]
Optimal Aperture	8.87
Optimal Annulus	20.26
Transit Duration (day)	0.1123 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.043 vs 0.1388 (deviation 0.1σ)
Duration fit vs published	0.1123 d vs 0.1046 d (deviation 1.31σ)
Mid-time O–C	-5.3 min
Depth SNR	3.3
Residual scatter	39.43 %

Light curve

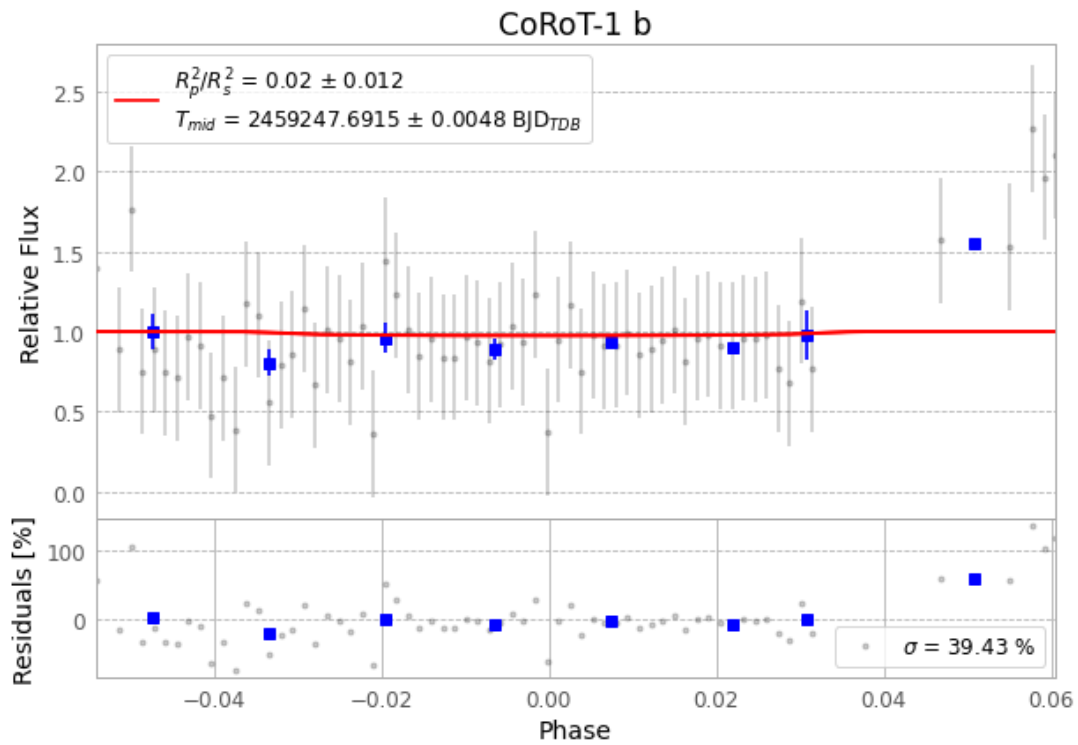


Figure 1 — FinalLightCurve_CoRoT-1 b_2021-02-01.png (SHA-256 95ba19df7c81d877...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [330, 271], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4d17635862692cde...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

89 FITS frames (58 MB), CoRoT-1210202023012.FITS → CoRoT-1210202065412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-210202.log (b892687b6d56...), FinalLightCurve_CoRoT-1 b_2021-02-01.png (95ba19df7c81...), FinalLightCurve_CoRoT-1 b_2021-02-01.csv (9a09ed039d33...)

Compute resources

Wall time 169.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 01:17Z.